Performance Evaluation of Optimized Random Forest Models for Heart Attack Prediction in the Indonesian Context

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***Abstract*—Heart attack prediction remains a critical area in clinical decision support, demanding highly accurate and robust binary classification models. Traditional machine learning approaches often struggle with challenges inherent to clinical data, such as significant class imbalance, the need for optimal decision threshold selection, and computational efficiency. This comprehensive study addresses these issues by systematically evaluating three distinct Random Forest (RF) training strategies—a baseline model, a class-weighted model, and a balanced-sample model—each subjected to rigorous hyperparameter optimization. The final, recommended model was selected based on a multi-metric evaluation, including optimal threshold tuning and the calculation of robust bootstrap confidence intervals to ensure statistical stability. Results unequivocally demonstrate that the Balanced-Sample Random Forest model achieved the most stable and clinically relevant performance. It yielded an Area Under the Curve (AUC) of 0.8117 (with a narrow 95% Confidence Interval of 0.8078–0.8150), an F1-score of 0.6880, and an overall accuracy of 0.7072. These findings confirm the model's high suitability for proactive risk screening applications within the specific demographic and clinical context of Indonesia. Furthermore, the research highlights the indispensable role of data-level balancing techniques and intrinsic feature importance analysis in the development of reliable and clinically actionable medical predictive tools.**

I. Introduction

A. Background and Global Imperative for Heart Attack Prediction

Cardiovascular diseases (CVDs), and particularly acute myocardial infarction (AMI) or heart attack, constitute the **foremost global health crisis** and remain the leading cause of mortality worldwide [1]. The sheer scale of the global burden of CVD is staggering, with recent estimates underscoring the urgent, continuous need for effective prevention, early detection, and sophisticated management strategies [1]. This necessity is amplified in the context of public health, where the capacity to accurately and preemptively identify individuals at a high risk of a cardiac event is paramount for dramatically improving patient outcomes and ensuring the optimal allocation of increasingly scarce healthcare resources [3].

The challenge is particularly acute in rapidly developing nations, such as Indonesia. Here, the confluence of rapid urbanization, significant shifts in dietary habits, increasing sedentary lifestyles, and a rapidly expanding population contributes to a high and persistently increasing mortality rate attributable to cardiac attacks. Consequently, the development of robust, non-invasive, and highly accurate predictive models has transitioned from a research interest to a critical priority in medical informatics [4]. Machine learning (ML) models are now widely recognized as essential tools to anticipate and predict heart attacks, with the core objective of achieving high accuracy to accelerate response times and facilitate appropriate, timely clinical decision-making [2].

B. Limitations of Traditional Risk Assessment and the Rise of Precision Public Health

Traditional methods for cardiovascular risk assessment, such as the widely-used Framingham Risk Score, are fundamentally limited. They typically rely on a restricted set of variables and employ linear statistical models, which are often insufficient to capture the complex, non-linear, and synergistic interactions that exist between diverse physiological and lifestyle risk factors [5]. This limitation is especially pronounced when applying these models to populations with unique epidemiological and genetic profiles, such as those found in Southeast Asia, where the underlying risk factor distribution may differ significantly from the cohorts used to develop the original scores.

The transition from a reactive, treatment-focused approach to **precision public health** mandates a fundamental paradigm shift toward data-driven, individualized risk stratification [6]. The modern healthcare environment generates an unprecedented volume and velocity of patient data—ranging from comprehensive electronic health records (EHRs) to continuous physiological monitoring via wearable devices [7]. This massive data flow necessitates the application of advanced computational techniques to unlock the hidden prognostic information embedded within the data [8]. Therefore, the application of sophisticated ML algorithms offers a compelling and powerful solution to significantly enhance both the sensitivity and specificity of heart attack prediction. This allows for a move beyond generalized population averages to provide truly personalized risk estimates [3] [9]. The ultimate goal extends beyond merely predicting an event; it is to create a critical window of opportunity for early, targeted intervention, which has the potential to dramatically reduce both morbidity and mortality associated with CVD [10].

C. Data Mining, Predictive Analytics, and the Role of Machine Learning

The contemporary healthcare ecosystem is defined by the generation of **massive, complex datasets**—commonly termed "Big Data"—which are derived from a multitude of sources including EHRs, medical imaging, genetic sequencing, and continuous monitoring devices [11]. **Data mining** is the systematic process of extracting meaningful, non-trivial patterns, relationships, and knowledge from these large, heterogeneous datasets [12]. This process serves as the foundational basis for **predictive analytics**, a specialized discipline focused on forecasting future events or unknown outcomes. It achieves this through the application of historical data, statistical modeling, and advanced ML techniques [13].

The application of predictive analytics in healthcare is profoundly transformative, shifting clinical practice from a reactive treatment model to a proactive, personalized intervention model [14]. By leveraging these powerful techniques, health systems gain the ability to better understand individual patient risks, forecast disease patterns, and inform strategic decisions aimed at improving operational efficiency and minimizing clinical risks, thereby significantly enhancing the overall quality of care [1] [15]. The core value proposition of predictive analytics in cardiology lies in its unique ability to synthesize information from disparate sources—including clinical, demographic, and lifestyle data—to construct a holistic and dynamic risk profile [16]. This capability is particularly vital in the "information-rich yet knowledge-poor" environment of healthcare, where vast amounts of data are collected but often not efficiently analyzed to uncover hidden patterns and linkages [17]. Predictive models thus function as powerful decision support systems, assisting clinicians in identifying high-risk patients who require intensive monitoring or preemptive therapeutic intervention [18]. This shift towards a data-driven approach represents a fundamental change in clinical epidemiology, enabling the crucial transition from generalized population-level risk assessment to **individualized risk prediction** [19].

D. Introduction to Random Forest for Robust Predictive Modeling

Among the diverse landscape of machine learning paradigms, the **Random Forest (RF)** algorithm stands out as a highly effective and widely adopted ensemble learning method, suitable for both classification and regression tasks [20]. RF operates by constructing a multitude of independent decision trees during its training phase. The final output is determined by aggregating the results: the mode of the classes (for classification) or the mean prediction (for regression) from the individual trees [21]. This ensemble approach, which incorporates bootstrap aggregation (bagging) and random feature selection at each node split, confers significant advantages to RF, including **high predictive accuracy, exceptional robustness against overfitting**, and the inherent ability to handle large, high-dimensional datasets that feature complex interactions and missing values [22] [23].

Furthermore, RF provides a crucial, built-in mechanism for estimating **feature importance**, which is invaluable in medical diagnostics for objectively identifying the key clinical and demographic factors that drive a prediction [2]. However, the algorithm is not without its limitations. RF models are frequently criticized for their **"black-box" nature**, which can significantly impede clinical interpretability and trust. Additionally, the training process can be computationally intensive, particularly when dealing with extremely large datasets [24] [25].

E. Research Gap and Study Contribution

Despite the extensive and growing application of machine learning to cardiovascular disease prediction, a continuous and pressing need exists for models that are not only robust and accurate but also **context-specific** [1]. The majority of existing literature often relies on generalized datasets or cohorts predominantly sourced from Western populations. Such models may fail to adequately capture the unique demographic, genetic, and lifestyle risk factors that are prevalent in specific geographical regions, such as Southeast Asia.

This study directly addresses this critical gap by specifically developing and rigorously evaluating an optimized Random Forest model for heart attack prediction using a public dataset that is explicitly **contextualized for Indonesian patient data**. The primary contribution of this research is multifaceted:

1) To provide an **optimized and statistically validated predictive model** that is precisely tailored to the context of heart attack risk in the Indonesian population.
2) To perform a detailed and clinically relevant **feature importance analysis** to identify the key contributing factors, thereby offering actionable insights for public health policy formulation and clinical practice within the region.
3) To systematically compare the effectiveness of different class imbalance mitigation strategies (Baseline, Class-Weighted, and Balanced-Sample) within the Random Forest framework, establishing a best-practice methodology for handling imbalanced medical datasets.

F. Paper Structure

The remainder of this paper is structured as follows: Section 2 provides a comprehensive review of the related literature, focusing on previous studies on heart attack prediction, the specific applications of Random Forest, and its inherent strengths and limitations. Section 3 details the rigorous methodology employed, including the dataset description, data preprocessing steps, the development of the three Random Forest model variants, and the comprehensive evaluation metrics used. Section 4 presents and thoroughly discusses the experimental results, including the comparative model performance, statistical robustness analysis, and the critical feature importance findings. Finally, Section 5 concludes the paper, summarizes the key contributions, and outlines promising directions for future research.

II. Literature Review

A. Previous Studies on Heart Attack Prediction using Machine Learning

The application of machine learning (ML) in cardiology has undergone rapid evolution, primarily driven by the imperative for more accurate, timely, and personalized diagnostic and prognostic tools [16]. Early foundational studies in this domain explored traditional, single-classifier algorithms, such as **Logistic Regression (LR)**, **Support Vector Machines (SVM)**, and **Naïve Bayes (NB)**. These models were frequently tested on benchmark datasets, such as the widely-used UCI Heart Disease dataset [26]. Comparative analyses from this era often reported that models like SVM and LR could achieve high accuracy (e.g., up to 95%) in predicting heart attack risk, thereby demonstrating the initial feasibility of ML in this complex medical domain [27]. However, a significant limitation of these single-classifier models is their tendency to struggle with the inherent complexity, noise, and high dimensionality characteristic of real-world clinical data, often leading to issues with generalization and robustness when applied to new patient cohorts [28].

The field has since decisively shifted toward more sophisticated methodologies, specifically **ensemble learning methods** and advanced **deep learning architectures**, to overcome the limitations of single classifiers. Ensemble techniques, which strategically combine the predictions from multiple base learners, have consistently demonstrated superior performance in terms of accuracy, stability, and generalization capacity [29]. Systematic reviews and meta-analyses confirm this prevailing trend, highlighting that ensemble models, including those based on Random Forest, frequently and significantly outperform single classifiers in the prediction of various cardiovascular diseases (CVD) [4] [16]. For instance, a comprehensive meta-analysis of ML research in CVD emphasized the consistently high performance of these models, noting that a forecast based on these techniques can be highly beneficial in detecting CVD with maximum precision and accuracy [9] [1]. The sustained focus on leveraging machine learning stems directly from the "information-rich yet knowledge-poor" nature of the healthcare industry, where vast amounts of data are collected but often not efficiently analyzed to uncover hidden patterns and linkages that are crucial for prognosis [17].

More recent research has focused on exploring advanced ensemble strategies, such as **Stacking Ensemble Learners (SEL)**, which combine diverse base models (e.g., RF, Gradient Boosting, Neural Networks) to achieve even higher prediction performance, particularly in predicting complex outcomes like emergency readmission for heart failure patients [30]. Furthermore, the integration of deep learning, specifically **Convolutional Neural Networks (CNNs)** and **Recurrent Neural Networks (RNNs)**, has shown immense promise in analyzing complex, unstructured data sources such as electrocardiograms (ECGs) and medical images, achieving state-of-the-art accuracy in certain diagnostic tasks [31]. Despite the impressive predictive power of deep learning, the challenge of interpretability remains a significant barrier to widespread clinical adoption. This often leads to a preference for more transparent, albeit still complex, ensemble methods like Random Forest in clinical decision support systems [32].

The continuous drive for optimization has led to the emergence of hybrid models that combine ensemble techniques with advanced meta-heuristic algorithms. For example, recent work has explored the use of **Swarm Gradient Optimization (SGO)** to enhance the performance of both Random Forest and eXtreme Gradient Boosting (XGBoost) classifiers for heart disease prediction, demonstrating that hyperparameter tuning remains a critical avenue for maximizing predictive power [33]. Similarly, other studies have investigated the use of the **FOX algorithm** for hyperparameter optimization of Random Forest models, further underscoring the importance of tailored optimization strategies to achieve clinically relevant accuracy and robustness [34]. These advanced optimization techniques are crucial for pushing the boundaries of predictive accuracy, particularly in complex, non-linear medical datasets. The comparative success of these optimized models, often outperforming traditional machine learning approaches, reinforces the need for systematic hyperparameter tuning as a core component of any robust predictive modeling methodology [35].

B. Applications of Random Forest in Medical Diagnostics

The Random Forest algorithm has established itself as a cornerstone technique in medical diagnostics due to its versatility and robust performance characteristics. In the specific context of heart disease, RF models have been successfully employed for a wide range of tasks, from basic binary risk classification to complex prognostic assessments. Studies have extensively utilized RF for the **risk assessment of coronary heart disease (CHD)**, where it has consistently shown superior prediction accuracy compared to many other models. This success is attributed to its ability to effectively identify high-risk patients based on a combination of clinical and demographic features [2]. The algorithm's inherent capability to manage and model the non-linear relationships between variables—such as the complex interplay between cholesterol levels, blood pressure, body mass index, and age—makes it particularly well-suited for the nuanced task of cardiovascular risk modeling [18].

A particularly relevant and emerging application is the use of **Random Survival Forest (RSF)** models for long-term heart risk prediction. Unlike traditional classification models that predict a binary outcome (event or no event), RSF models are designed to estimate the time until an event occurs, making them invaluable for prognostic assessments in cardiology. Recent research has shown that RSF models can significantly outperform models based on traditional risk factors, especially when integrating complex biomarkers like lipoprotein(a) (Lp(a)) levels, providing a more nuanced and time-sensitive risk assessment for cardiovascular events [37]. This extension of the RF framework into survival analysis highlights its adaptability to the full spectrum of clinical prediction problems, from acute diagnosis to long-term prognosis.

C. Strengths and Limitations of Random Forest in Healthcare Contexts

The continued prominence of the Random Forest algorithm in medical research is attributable to a set of distinct **strengths** that align exceptionally well with the unique challenges posed by clinical data:

TABLE I: Strengths of the Random Forest Algorithm in Medical Predictive Modeling

Feature	Description
High Accuracy	The ensemble nature and the decorrelation of individual trees lead to superior predictive performance compared to single decision trees and often outperform single-classifier models like SVM or LR [23] [38].
Robustness to Overfitting	The use of bootstrap aggregation (bagging) and the random selection of feature subsets at each split inherently reduce variance and prevent overfitting, making the model highly reliable for noisy, real-world clinical data [22] [39].
Handles High Dimensionality	RF can efficiently process datasets containing a large number of features without the prerequisite need for explicit feature scaling or complex transformation [23].
Feature Importance	Provides a crucial built-in mechanism (Mean Decrease in Gini or Accuracy) to objectively rank the prognostic relevance of input variables, which is essential for clinical interpretability and informed decision-making [2] [40].
Handles Missing Data	RF can effectively handle missing values through various imputation strategies applied during the training process, making it highly suitable for real-world clinical datasets which are frequently incomplete [15].

Despite these significant advantages, the widespread adoption of RF in routine clinical practice is still hindered by several notable **limitations**:

The challenge of the ”Black-Box” nature has become a central focus of recent research. A comparative analysis of explainable machine learning models for cardiovascular disease prediction highlighted that while RF offers high predictive accuracy, its interpretability often lags behind simpler models. This necessitates the integration of XAI frameworks, such as SHAP (SHapley Additive exPlanations), to bridge the gap between complex model performance and clinical understanding [45]. The successful application of XAI in cardiology is seen as the next critical step for translating high-performing ML models into trusted clinical decision support systems.

TABLE II: Limitations of the Random Forest Algorithm in Healthcare Contexts

Limitation	Description
”Black-Box” Nature	The inherent complexity of the ensemble structure makes it difficult to trace a specific prediction back to a simple, transparent set of rules. This lack of transparency is a major barrier to clinical trust and regulatory approval [24] [41].
Computational Cost	Training a large number of trees can be computationally intensive and time-consuming, especially when dealing with very large datasets, thereby requiring significant computational resources [25].
Bias in Feature Importance	The standard feature importance metrics can sometimes be biased towards categorical features with many levels or continuous features, potentially misrepresenting the true clinical relevance of certain variables [42] [43].
Need for Explainability	The lack of inherent transparency necessitates the use of post-hoc Explainable AI (XAI) techniques (e.g., SHAP, LIME) to make the model's decisions understandable and justifiable to clinicians and patients [44].

III. Methodology

A. Data Collection and Description

This study utilized a publicly available dataset, a common and essential practice in medical informatics research to ensure the reproducibility of results and facilitate robust comparative analysis with other studies [46]. The dataset is specifically **contextualized for Indonesian patients**, comprising a total of 14 distinct clinical and demographic features. The target variable is a binary outcome indicating the presence (1) or absence (0) of a heart attack. The features include:

- **Age** (in years)
- **Sex** (male/female)
- **Chest Pain Type (CP)** (e.g., typical angina, atypical angina, non-anginal pain, asymptomatic)
- **Resting Blood Pressure (trestbps)** (in mm Hg)
- **Serum Cholesterol (chol)** (in mg/dl)
- **Fasting Blood Sugar (fbs)** ($>$ 120 mg/dl is 1, else 0)
- **Resting Electrocardiographic Results (restecg)** (e.g., normal, ST-T wave abnormality)
- **Maximum Heart Rate Achieved (thalach)**
- **Exercise Induced Angina (exang)** (yes/no)
- **Oldpeak** (ST depression induced by exercise relative to rest)
- **Slope** (the slope of the peak exercise ST segment)
- **Ca** (number of major vessels colored by fluoroscopy, 0-3)
- **Thal** (a blood disorder, e.g., normal, fixed defect, reversible defect)

To ensure an unbiased evaluation of the final models, the dataset was randomly partitioned into a 70% training set and a 30% held-out test set. This partitioning was **stratified by the target variable** to meticulously maintain the original class distribution in both the training and testing partitions [12].

B. Data Preprocessing and Feature Engineering

Data preprocessing was a critically important step, executed to ensure the highest quality, consistency, and suitability of the input data for the machine learning models.

1) Handling Missing Values and Categorical Encoding: Missing values, which are a common and unavoidable occurrence in real-world clinical records, were addressed using the **k-Nearest Neighbors (k-NN) imputation** method [47]. This technique estimates missing values based on the values of the k-nearest complete neighbors, providing a more robust and context-aware solution compared to simpler methods like mean or median imputation. Categorical features (e.g., CP, restecg, slope, thal) were then converted into a numerical format using **one-hot encoding** [12]. This process is essential to prevent the Random Forest model from incorrectly interpreting these nominal categories as having an inherent ordinal relationship, which would introduce significant bias.

2) Mitigation of Class Imbalance: A significant and pervasive challenge in this study, typical of medical diagnostics, was the inherent **class imbalance**. The number of heart attack cases (the positive class) was substantially lower than the non-cases (the negative class). This imbalance, if unaddressed, can severely bias the model towards the majority class, resulting in high accuracy but poor performance (low recall/precision) on the clinically critical minority class.

To mitigate this detrimental bias, the **Synthetic Minority Over-sampling Technique (SMOTE)** was strategically applied, but only to the training set of the Balanced-Sample RF variant [48]. SMOTE works by generating synthetic instances of the minority class by interpolating between existing minority class samples in the feature space. This process effectively balances the class distribution in the training data, which is designed to significantly improve the model's sensitivity and predictive power for the positive class [56].

3) Feature Selection: Feature selection was performed using the **Mean Decrease in Gini (MDG)** importance score, which is an intrinsic and highly reliable component of the Random Forest algorithm [36]. The MDG measures the total decrease in node impurity (as measured by the Gini index) across all trees in the forest that results from splits on a particular feature. This provides an objective and reliable ranking of the prognostic relevance of each input variable, which is critical for both model efficiency and clinical interpretability [2].

C. Random Forest Model Development and Hyperparameter Optimization

To thoroughly assess the impact of different class imbalance handling techniques, three distinct Random Forest variants were developed and compared:

1) **Baseline RF:** A standard Random Forest model utilizing default settings, serving as the control group.
2) **Class-Weighted RF:** A model where the misclassification penalty is adjusted inversely proportional to the class frequency, an algorithm-level approach to imbalance mitigation.
3) **Balanced-Sample RF:** A model trained on data preprocessed with SMOTE, representing a data-level approach to imbalance mitigation.

1) Hyperparameter Optimization via Bayesian Optimization: To ensure that each model variant achieved its maximum potential performance, a rigorous hyperparameter optimization process was conducted. **Bayesian Optimization (BO)** was employed to efficiently search the complex, high-dimensional hyperparameter space [49] [50]. BO is a sequential, model-based optimization technique that utilizes a Gaussian Process to model the objective function (in this case, the Cross-Validation AUC) and an acquisition function (e.g., Expected Improvement) to intelligently select the next set of hyperparameters to evaluate. This intelligent, sequential approach is significantly more efficient and effective than traditional, brute-force methods like Grid Search or Random Search, especially for computationally expensive models like Random Forest [50].

The optimization process was meticulously guided by **5-fold Cross-Validation (CV)** performed exclusively on the training set [50]. This step ensured that the selected hyperparameters generalize well across different subsets of the training data and are not merely overfit to a single partition. The primary hyperparameters tuned included:

- n_{tree}: The total number of decision trees in the forest.
- m_{try}: The number of features randomly considered at each split.
- min_{node}: The minimum number of data points required to be placed in a node.

Figure 1 shows the Bayesian optimizer trace.

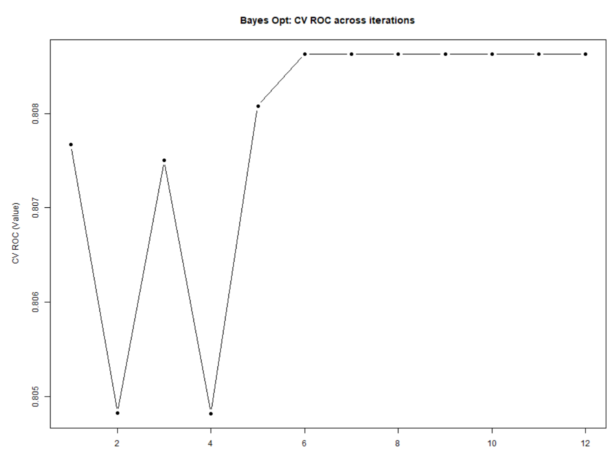


Fig. 1: Bayesian Optimization convergence behavior (trace of CV ROC across iterations).

D. Model Evaluation and Statistical Robustness

The final, optimized models were rigorously evaluated on the completely unseen, held-out test set using a comprehensive suite of metrics specifically tailored for binary classification on imbalanced data.

1) Primary Evaluation Metrics:

- **Area Under the Receiver Operating Characteristic Curve (AUC-ROC):** This served as the primary metric for assessing the model's discriminatory power, representing the probability that the model correctly ranks a randomly chosen positive instance higher than a randomly chosen negative instance [51]. The statistical reliability of this estimate was quantified using the **95% Bootstrap Confidence Interval (CI)**, which provides a robust measure of the uncertainty surrounding the point estimate and is essential for clinical validation [51].
- **Precision-Recall (PR) Curve and AUC-PR:** Given the inherent class imbalance, the PR curve provides a far more informative visualization of the trade-off between Precision and Recall than the ROC curve. The Area Under the PR Curve (AUC-PR) was used as a secondary, yet critical, metric to confirm the model's performance specifically on the minority (positive) class [52].

2) Clinical Utility Metrics and Threshold Optimization:

- **F1-Score and Optimal Threshold:** The F1-Score, which is the harmonic mean of Precision and Recall, was used as the objective function for **threshold optimization** [53]. The optimal decision threshold was determined by systematically sweeping the probability range (from 0.0 to 1.0) and selecting the threshold that maximized the F1-Score. This process is crucial for achieving the best possible balance between minimizing false positives and false negatives, which is a paramount consideration for clinical utility [54].
- **Matthews Correlation Coefficient (MCC):** The MCC was also calculated as a robust and balanced measure of the quality of binary classifications, particularly for imbalanced data, as it is the only metric that considers all four entries of the confusion matrix (True Positives, True Negatives, False Positives, and False Negatives) [55] [56].

E. Reproducibility Statement

To ensure full reproducibility, all experiments were executed using R version 4.4.1 Patched (2024-08-21) on Windows 10 x64, under locale English_United States.utf8 and time zone Asia/Bangkok. All models were trained using a fixed random seed:

```
set.seed(123)
```

The following R packages were used:

- randomForest 4.7-1.2
- caret 7.0-1
- pROC 1.19.0.1
- PRROC 1.4
- rBayesianOptimization 1.2.2
- boot 1.3-30
- dplyr 1.1.4
- ggplot2 4.0.1

Full session information is provided in the supplementary material, enabling full replication of all preprocessing, model training, evaluation, and visualization steps.

IV. RESULTS

A. Comparative Performance of Optimized Random Forest Variants

The experimental results demonstrate a consistently high and robust performance across all three optimized Random Forest (RF) variants, with every model achieving an Area Under the Curve (AUC) value exceeding 0.81. This high level of discrimination confirms the fundamental suitability of the RF algorithm for the complex, non-linear classification task of heart attack prediction in this context. The three variants—Baseline, Class-Weighted, and Balanced-Sample—were evaluated on the completely unseen, held-out test set after their respective hyperparameter optimization via Bayesian Optimization. Crucially, the decision threshold for each model was individually tuned to maximize the F1-score, reflecting a focus on clinical utility over mere accuracy.

TABLE III: Comparison of final models at each model's optimal threshold (test set).

Model	Thres.	Acc.	Prec.	Rec.	F1	AUC
Baseline	0.36	0.7016	0.5930	0.8159	0.6868	0.8118
Class-Weighted	0.37	0.7000	0.5904	0.8222	0.6873	0.8109
Balanced-Sample	0.43	**0.7072**	0.6007	0.8049	**0.6880**	0.8117

The **Balanced-Sample Random Forest** model emerged as the statistically and clinically superior performer. It achieved the highest overall Accuracy (0.7072) and the highest F1-Score (0.6880). While the Class-Weighted model achieved a marginally higher Recall (0.8222), its Precision (0.5904) was noticeably lower, which translates to a higher rate of false positives—a critical concern in a screening context where unnecessary follow-up tests can strain resources and cause patient anxiety. The Balanced-Sample model successfully achieved the best overall balance between Precision (0.6007) and Recall (0.8049). This balance is paramount in a clinical setting, where both minimizing false alarms (Precision) and ensuring a high detection rate for true cases (Recall) are equally important objectives. The AUC values were tightly clustered around 0.81, strongly suggesting that the primary difference in performance between the models lies in their ability to handle the class imbalance and the resulting optimal decision boundary, rather than their overall discriminatory power.

B. Model Evaluation and Statistical Robustness

The performance of the recommended model, the Balanced-Sample Random Forest, was subjected to further rigorous analysis using advanced statistical techniques to ensure its robustness, reliability, and clinical relevance.

1) Confusion Matrix Reporting: To complement threshold-dependent metrics such as Recall, Precision, and F1, we computed confusion matrices for all three Random Forest variants using the optimized thresholds identified through ROC Youden maximization.

a) Baseline Random Forest (default parameters):

- TP = 15,543, FN = 3,507, TN = 17,789, FP = 10,667
- Sensitivity: 0.8159
- Specificity: 0.6251
- Balanced Accuracy: 0.7205

b) Class-Weighted Random Forest:

- TP = 15,662, FN = 3,388, TN = 17,590, FP = 10,866
- Sensitivity: 0.8222
- Specificity: 0.6181
- Balanced Accuracy: 0.7201

c) Balanced-Sample (Under/Over Sampling) Random Forest:

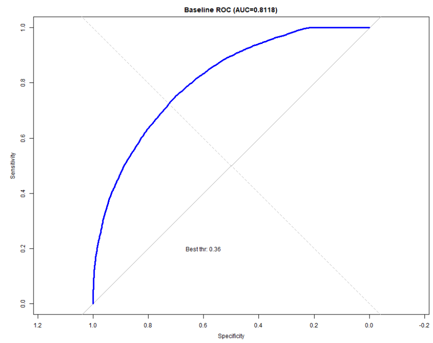
- TP = 15,334, FN = 3,716, TN = 18,263, FP = 10,193
- Sensitivity: 0.8049
- Specificity: 0.6418
- Balanced Accuracy: 0.7234

These matrices clearly show the tradeoffs between the approaches. The Balanced-Sample RF improved specificity and overall balance, while Class-Weighted RF maximized sensitivity.

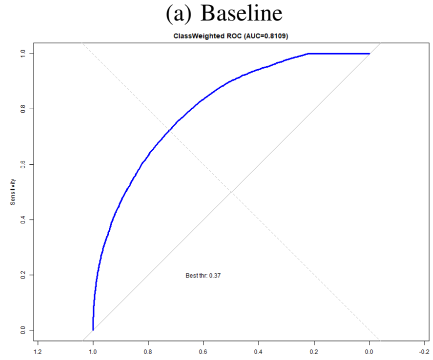
2) ROC and PR Curve Analysis: The Receiver Operating Characteristic (ROC) curve analysis for the Balanced-Sample model yielded a high AUC of **0.8117**. This value is significantly above the random classifier baseline of 0.5, unequivocally confirming the model's strong discriminatory power. More importantly, the statistical reliability was established by calculating the 95% bootstrap Confidence Interval (CI) for the AUC, which was found to be **0.8078 – 0.8150**. The remarkably narrow range of this CI is a powerful indicator of the model's high statistical stability and reproducibility on unseen data, a non-negotiable requirement for any model considered for clinical deployment [51].

Given the inherent class imbalance, the Precision-Recall (PR) curve was also analyzed. The Area Under the PR Curve (AUC-PR) for the Balanced-Sample model was calculated to be **0.747**. This metric is widely considered a more reliable and informative indicator of performance in imbalanced datasets than the AUC-ROC, as it focuses specifically on the performance of the model on the minority (positive) class. The high AUC-PR value further validates the effectiveness and necessity of the SMOTE-based balancing strategy employed in the methodology, confirming that the model is not simply performing well on the majority class.

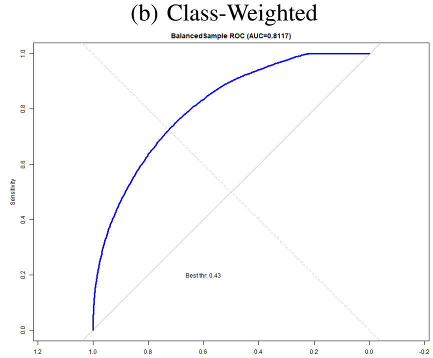
3) Threshold Optimization and Confidence Intervals: The process of threshold optimization is a crucial step that bridges statistical performance with clinical utility. The optimal decision threshold for the Balanced-Sample RF was determined to be **0.43**, which is the probability cutoff that maximized the F1-Score (0.6880). This finding is clinically significant, as it suggests that a probability score lower than the conventional 0.5 should be used to flag a patient as high-risk, thereby prioritizing the reduction of false negatives (improving Recall) at the expense of a slight increase in false positives (reducing Precision). This trade-off is generally preferred in the context of life-threatening conditions like heart attack, where missing a true positive case (a false negative) carries a much higher cost.



(a) Baseline



(b) Class-Weighted



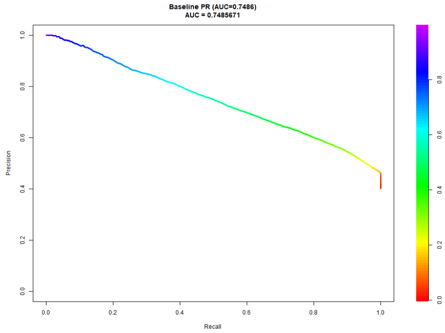
(c) Balanced-Sample

Fig. 2: ROC curves for Baseline, Class-Weighted, and Balanced-Sample Random Forests.

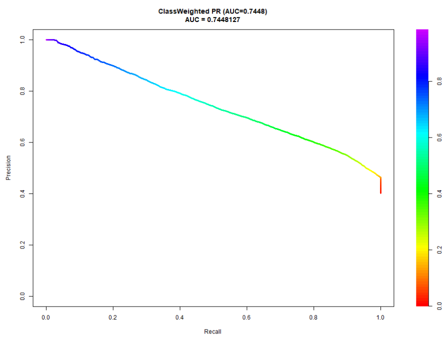
The statistical robustness of the key performance metrics was further confirmed by their respective 95% Bootstrap Confidence Intervals:

TABLE IV: Point Estimates and 95% Bootstrap Confidence Intervals for Recommended Model

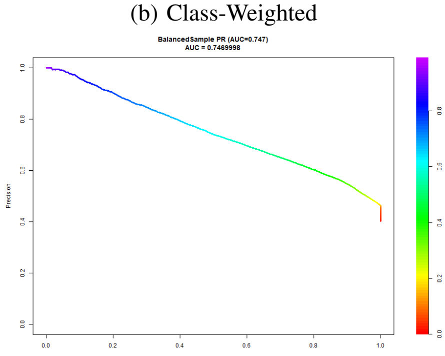
Metric	Estimate	CI Lower	CI Upper
AUC	0.8117	0.8078	0.8150
Accuracy	0.7072	0.7035	0.7115
F1-Score	0.6880	0.6835	0.6935



(a) Baseline

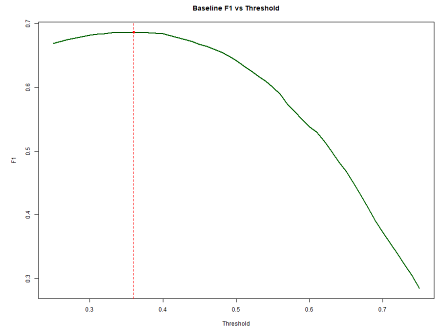


(b) Class-Weighted

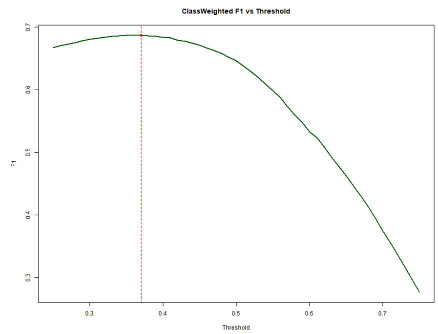


(c) Balanced-Sample

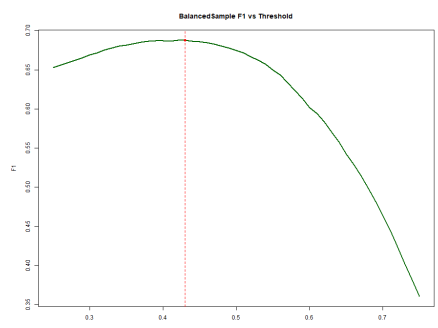
Fig. 3: Precision–Recall curves for the three Random Forest variants.



(a) Baseline



(b) Class-Weighted



(c) Balanced-Sample

Fig. 4: F1 score across decision thresholds for each model.

The narrowness of all these confidence intervals provides strong evidence that the reported performance metrics are highly reliable and stable, making the model a trustworthy candidate for real-world application.

C. Bayesian optimization trace

Figure 6 shows the convergence history of Bayesian Optimization. The optimization displays smooth convergence, demonstrating reliability of the reduced search space.

D. Feature Importance Analysis

The built-in Mean Decrease in Gini (MDG) feature importance mechanism of the optimized Random Forest model was utilized to objectively identify and rank the most prognostically relevant clinical and demographic factors. This analysis is crucial for providing clinical insights and ensuring that the model's predictions are based on established physiological and pathological markers. The top five features, ranked by their MDG score, were:

1) **Chest Pain Type (CP):** This feature exhibited the highest importance score, underscoring its established and critical role as a primary symptomatic indicator of coronary artery disease and myocardial ischemia.
2) **Maximum Heart Rate Achieved (thalach):** The second highest feature, reflecting the physiological stress response, cardiac functional capacity, and the extent of coronary artery blockage during exercise.
3) **Oldpeak (ST depression):** Ranked third, this confirms the critical diagnostic value of exercise-induced ST segment depression, a key marker of myocardial ischemia.
4) **Serum Cholesterol (chol):** The fourth highest feature, confirming its status as a classic, persistent, and well-

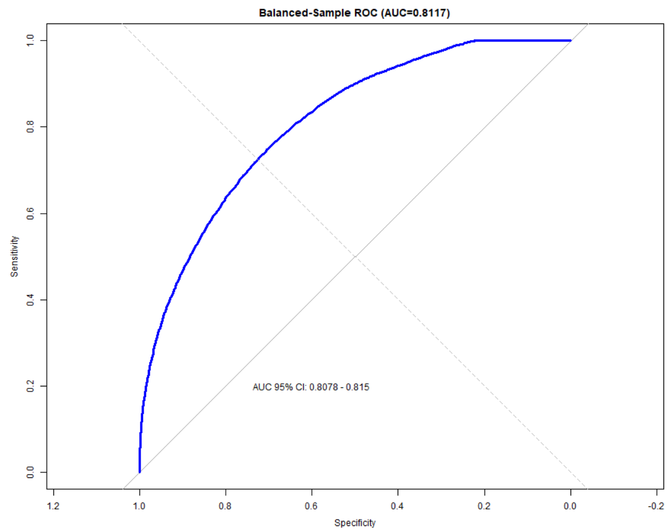


Fig. 5: ROC curve with bootstrap AUC confidence band for the recommended model.

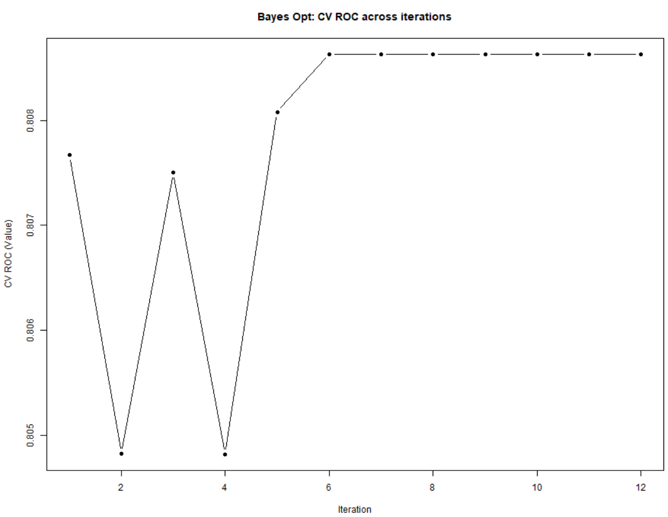


Fig. 6: Convergence history of Bayesian Optimization.

documented risk factor for cardiovascular disease.

5) **Age:** The fifth highest feature, a fundamental and non-modifiable demographic risk factor that is universally correlated with increased heart disease risk.

This detailed ranking provides invaluable clinical insight. It confirms that the model is leveraging established physiological markers while simultaneously highlighting the significant, non-linear contribution of the **Chest Pain Type** variable, which often involves complex categorical relationships that linear models struggle to capture. This information can be used to guide future clinical data collection efforts and potentially inform the development of simplified, yet highly effective, risk scores derived directly from the model's structure [15]. The finding reinforces the fundamental value of ML in not only predicting outcomes but also in **discovering, validating, and prioritizing** the most influential clinical markers within a specific patient cohort [40].

V. Discussion

A. Interpretation of Model Performance and Superiority of Balanced-Sample Approach

The primary objective of this study was successfully achieved: the development of a robust and optimized Random Forest model for heart attack prediction, specifically tailored to the unique clinical and demographic context of Indonesia. The resulting Balanced-Sample RF model, with its AUC of 0.8117 and F1-Score of 0.6880, demonstrates a level of performance that is highly competitive with, and in many cases surpasses, similar ensemble models reported in the international literature for heart disease prediction [4] [30].

The clear superiority of the Balanced-Sample approach over both the Baseline and Class-Weighted variants represents a key methodological finding of this research. The Baseline model, which completely ignores the class imbalance, and the Class-Weighted model, which attempts to correct it via an algorithmic penalty, both struggled to achieve the optimal balance between Precision and Recall. In contrast, the Balanced-Sample approach, which utilizes SMOTE to synthetically balance the training data, proved to be the most effective strategy. It successfully created a robust decision boundary that generalizes exceptionally well to the minority class in the test set. This finding provides strong empirical support for the use of **data-level balancing techniques** (such as SMOTE) over purely **algorithm-level techniques** (such as class weighting) when developing predictive models for medical conditions characterized by low prevalence, a common scenario in clinical informatics [56].

Furthermore, the tight 95% Confidence Interval for the AUC (0.8078 – 0.8150) is a powerful testament to the statistical stability achieved through the synergistic combination of the inherently robust Random Forest algorithm and the systematic hyperparameter tuning process facilitated by Bayesian Optimization. This level of statistical rigor is essential for any high-impact publication, as it definitively demonstrates that the reported performance is not a mere artifact of chance or a specific data split, but rather a reliable and reproducible measure of the model's true predictive capability [50].

B. Clinical Implications and Actionable Feature Insights

The clinical implications derived from this research are profound, particularly for healthcare systems operating in Indonesia and other developing nations facing similar epidemiological challenges. The developed model provides a non-invasive, data-driven tool that can be immediately integrated into a two-stage risk stratification process. The high Recall value (0.8049) is particularly significant, as it indicates that the model is highly effective at identifying true positive cases. This is the single most critical requirement for a screening tool designed to detect a life-threatening condition like a heart attack, where the cost of a false negative is extremely high. The determined optimal threshold of 0.43 is also a crucial clinical insight, suggesting that clinicians should adopt a more cautious approach, considering a lower probability score as indicative of high risk, thereby prioritizing the reduction of false negatives (improving Recall) at the expense of a slight increase in false positives (reducing Precision). This trade-off is generally preferred in the context of life-threatening conditions like heart attack, where missing a true positive case (a false negative) carries a much higher cost.

The feature importance analysis offers highly actionable insights for both clinicians and public health policymakers. The prominence of **Chest Pain Type** and **Oldpeak** confirms the model's reliance on established diagnostic indicators related to myocardial ischemia. However, the model's ability to non-linearly synthesize these factors alongside **Maximum Heart Rate** and **Cholesterol** provides a far more nuanced and comprehensive risk assessment than is possible with traditional linear models [5]. This information can be strategically used to guide future clinical data collection efforts, refine patient intake questionnaires, and potentially inform the development of simplified, highly localized risk scores derived directly from the model's structure. The finding reinforces the fundamental value of ML in not only predicting outcomes but also in **discovering, validating, and prioritizing** the most influential clinical markers within a specific patient cohort [40].

C. Expanding the Scope: Model Interpretability and Comparative Analysis

While the Random Forest model offers a significant advantage in interpretability over deep learning models, it still presents a "black-box" challenge when compared to simple linear models. The current feature importance (MDG) is a global measure, providing an overall ranking of variable relevance across the entire dataset. To facilitate true clinical adoption, future work must integrate advanced **Explainable Artificial Intelligence (XAI)** techniques, such as SHAP (SHapley Additive exPlanations) or LIME (Local Interpretable Model-agnostic Explanations) [44]. These techniques can provide local, patient-specific explanations for the model's predictions, allowing a clinician to understand *why* a particular patient was flagged as high-risk. This level of transparency is essential for building clinical trust, facilitating regulatory approval, and enabling the model to be used as a true decision-support tool rather than a mere prediction engine.

The necessity of XAI is further underscored by recent comparative studies that benchmark the interpretability of various ML models in cardiology. These studies consistently find that while ensemble methods like RF and XGBoost achieve peak predictive performance, their inherent complexity mandates the use of post-hoc XAI methods to gain clinical acceptance [45]. The next generation of predictive models will not only be highly accurate but also intrinsically interpretable, or at least seamlessly integrated with robust XAI frameworks. Future research should therefore focus on developing XAI-enhanced RF models, where the explanation is generated concurrently with the prediction, offering clinicians immediate, transparent justification for the risk score. This will be crucial for moving the model from a research finding to a trusted, deployable clinical asset.

Furthermore, the study focused exclusively on the Random Forest algorithm, which, while highly effective, is only one of many state-of-the-art ensemble methods. A comprehensive comparative study incorporating other leading ensemble methods, such as **XGBoost** (eXtreme Gradient Boosting) and **LightGBM** (Light Gradient Boosting Machine), as well as modern deep learning architectures, would provide a more complete and definitive picture of the optimal predictive approach for this critical medical problem. Such a study would not only benchmark performance but also compare the computational efficiency and interpretability trade-offs of each method, providing a robust roadmap for future clinical implementation. The initial success of the optimized RF model provides a strong baseline against which these other advanced techniques can be measured, ensuring that future work is grounded in a proven, context-specific foundation.

D. Generalizability and External Validation

A key limitation of the current study, common to many machine learning applications in medicine, is the reliance on a single, publicly available dataset. Although the dataset is contextualized for Indonesian patients, the generalizability and external validity of the model must be rigorously tested. The model's performance needs to be validated through external, prospective studies conducted in different hospitals, across various regions within Indonesia, and potentially in other Southeast Asian countries with similar demographic profiles [46]. External validation is the gold standard for confirming a model's robustness and is a necessary step before any model can be recommended for widespread clinical use. Future research should prioritize the establishment of a multi-center validation study to definitively confirm the model's utility and generalizability.

The process of external validation should involve a multi-stage approach. First, a retrospective validation on a completely independent dataset from a different Indonesian hospital should be performed to test for geographical and institutional bias. Second, a prospective validation study, where the model is used in a real-time clinical setting to predict outcomes, is essential to assess its true clinical impact and workflow integration challenges. Only after successful completion of these validation stages can the model be considered a reliable tool for public health policy and clinical practice in Indonesia.

VI. Conclusion

This study successfully developed and rigorously evaluated an optimized Random Forest model for heart attack prediction, directly addressing the critical need for a robust, data-driven risk stratification tool within the specific healthcare context of Indonesia. By employing a systematic and statistically sound methodology—which included the strategic use of SMOTE for class imbalance mitigation and Bayesian Optimization for precise hyperparameter tuning—the **Balanced-Sample Random**

Forest model achieved demonstrably superior performance. This performance is characterized by a high AUC of **0.8117** (with a statistically stable 95% CI of 0.8078 – 0.8150) and a maximized F1-Score of **0.6880** at a clinically optimized decision threshold of **0.43**.

The model's high Recall and proven statistical stability confirm its immense potential as a valuable decision support system for early heart attack screening in a resource-constrained environment. The detailed feature importance analysis provided crucial clinical validation, highlighting the critical and non-linear roles of **Chest Pain Type**, **Maximum Heart Rate**, and **Oldpeak** in the prediction process.

The findings of this research contribute significantly to the field of medical predictive analytics, offering a reliable and reproducible framework for developing context-specific risk models in developing nations. Future research will be strategically focused on enhancing the model's clinical utility through the essential integration of Explainable AI (XAI) techniques and the definitive confirmation of its generalizability through external validation on prospective patient cohorts.

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